

Parssa Alimadad

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EDUCATION

University of California, Davis

Sep. 2025 – Jan. 2027

M.S. Electrical Engineering - GPA: 4.0

Highlights: IC Design & Tapeout, MOS Analog Circuit Design, THz & mm-Wave IC Design, EEC 100 - Circuits II, EEC 146A - IC Fabrication & EEC 1 - ECE Emerge Graduate Lab TA, IEEE UC Davis Officer

University of California, Davis

Sep. 2021 – Jun. 2025

B.S. Electrical Engineering - EEC 1 and EEC 111 Undergraduate Lab TA

TECHNICAL SKILLS

Electrical Debugging and Tools: DMM, Function Generator, Oscilloscope, Spectrum Analyzer, Power Supply, Soldering (SMD & THT), 3D Printing, Probe Station, Wafer Profilometer, Mask Aligner

Software: Cadence Virtuoso, Spectre, Calibre, Altium Designer, KiCAD, OrCAD Capture, LTSpice, PSpice, ModelSim, Intel Quartus FPGA, Arduino, Visual Studio, Silvaco CAD, ANSYS Lumerical, AutoCAD Inventor

Programming: C/C++, Java, Python, MATLAB, GPIO, Qiskit, Verilog HDL, Data Structures & Algorithms, LaTeX

WORK AND RESEARCH EXPERIENCE

Texas Instruments (Embedded Processing)

Jun. 2025 – Sep. 2025

Applications Electrical Engineering Intern

Dallas, TX

- Designed and implemented 8-layer mixed-signal TEM cell PCB on **Altium Designer** for TI's audio processing **microcontrollers** (AM275), optimized for rigorous **EMI compliance** testing
- Produced detailed electrical & logic **schematics** streamlining **mixed-signal PCB layout and fabrication**
- Sourced new and replacement components for board design; **cross-collaborated** with senior engineers on design shortcomings and implementing improvements
- Ensured stringent design conditions were met; minimizing **cross-talk** and **interference** from analog components

Applied Micro/Nano Electromagnetics Research Laboratory

Sep. 2023 – Present

THz Laboratory Graduate Researcher

Davis, CA

- Designing **sensing CMOS IC** for cancer detection through shift in electromagnetic response of live bio-samples
- Characterizing transmission & absorption of metamaterial properties using **THz time-domain spectroscopy**, **FT-IR**, and calibrated optical setups
- Automating experimental measurements and signal processing in **Python/MATLAB** for quantitative comparison of sample responses
- Fabricating novel **nanoscale** metamaterial structures and MEMS devices in the **UC Davis CNM2 cleanroom** and **UC Berkeley Marvell Nanofab** for tunable sensing applications

PROJECTS

200 MHz CMOS IQ Demodulator IC Design & Tapeout - TSMC 180nm CMOS PDK

Phase: 1 of 3

- Designed a 200 MHz zero-IF IQ demodulator in TSMC 180nm CMOS using **Cadence Virtuoso**, differential amplification, **buffer clocks**, quadrature LO generation, and mixers
- Performed block- and system-level **Spectre** simulations to evaluate gain, bandwidth, noise, & LO feedthrough
- Optimized CMOS sizing & biasing for low noise & power; maintained I/Q phase, amplitude, & **timing alignment**
- Completed custom layout, **DRC/LVS/PEX** verification, analyzing **parasitics** and **post-layout performance**
- Physical IC functionality to be tested on **custom PCB** after tapeout (ETA: Oct. 2026)

mm-Wave CMOS Power Amplifier - FreePDK 45nm

- Designed **multi-stage amplifier** in Cadence Virtuoso, achieving 40.4 dB @ 200 GHz with 10 GHz bandwidth
- Optimized transistor sizing, biasing conditions, and matching; verifying **low noise and power** achieving 270 uW
- Enabled design to be easily cascade-able, matching input and output to 50 ohms

THz Capstone Design Project - "Tunable and Reconfigurable Terahertz Technologies"

- Fabricated a time-tunable THz device using centrosymmetric perovskite thin film oxides in **clean room**
- Validated theoretical analysis of electromagnetic properties of fabricated material in **MATLAB/COMSOL**
- Awarded top outstanding ECE Senior Design Project; placed **Top 3 out of 173** in Sandia Design Competition